ROTEM Familiarisation for New Starters Ysbyty Gwynedd

Introduction

As part of the Obs Cymru project, a ROTEM Sigma machine (***Figure 1***) is available at Ysbyty Gwynedd.



Figure 1: ROTEM Sigma

Before reading on, watch this Introduction video:
https://www.youtube.com/watch?v=Aq57mEY7PyQ

Why use it?

The use of ROTEM thromboelastography, facilitates targeted, individualised blood product administration. This would otherwise be empirical using standard lab based blood tests.

When do I use it?

1. As part of the **Obs Cymru Project**: For all cases of measured blood loss> 1000ml
2. Non-Obstetric: Massive haemorrhage in theatres, critical care and emergency department.

Where is it?

The ROTEM Sigma machine is located in Maternity theatres.
Cartridges for Obstetric patients are located with the machine.
Cartridges for non-Obstetric patients are located in main theatres (next to ODP office).

How do I use it?

- ROTEM must be used as part of a Major Haemorrhage protocol.
- Fill a clotting blood test bottle with 4.5ml of blood. Place a patient sticker on the bottle with the bar code visible.
- Ask an ODP (or midwife) if they are available to process the sample with ROTEM.
- Following processing, view the results of the ROTEM test remotely using the application, **Secure Viewer** (See Appendix 3)**.**

Watch this short video on how to run a test yourself.
https://www.youtube.com/watch?v=hmJDaXH6Rfk

How do I interpret the results?

*All you need is: **1** time value, **1** amplitude value and **2** ways to activate clot. That's it!*

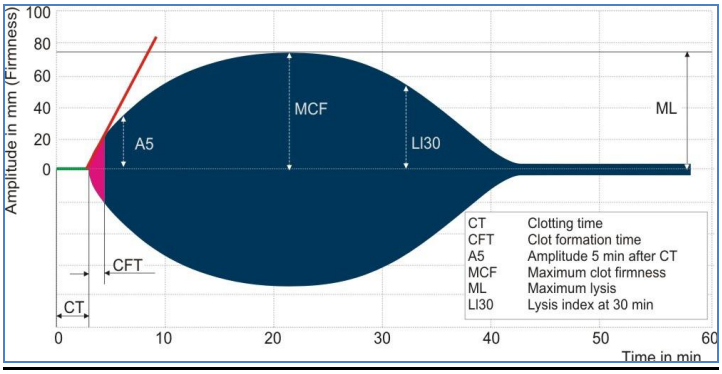


Figure 2:ROTEM Values

Values: (*Figure2*)

- **CT=Clotting Time**. It measures how long it takes for the clotting to be initiated (2mm amplitude). It is primarily affected by the levels of clotting factors present.
- **A5= Clot amplitude at 5 minutes after the clotting time**. It accurately predicts the final strength of the clot which is primarily determined by fibrinogen and platelet interaction.

Assays:

- **EXTEM**: by activating clot formation with thromboplastin (tissue factor), EXTEM gives an overview of the extrinsic pathway (factors VII, X, V, II, I), platelets and fibrinolysis.
- **FIBTEM**: differs from EXTEM by addition of a platelet-blocking agent. Consequently, FIBTEM looks at fibrinogen and fibrin polymerisation activity.

In order to determine what the patient requires, we only need to look at 3 values (***Figure 3):***

- **EXTEM CT**,
- **EXTEM A5** and the
- **FIBTEM A5.**

Detailed interpretation of results is not necessary. Just follow the algorithms within the major haemorrhage protocols (see appendix 1 &2).

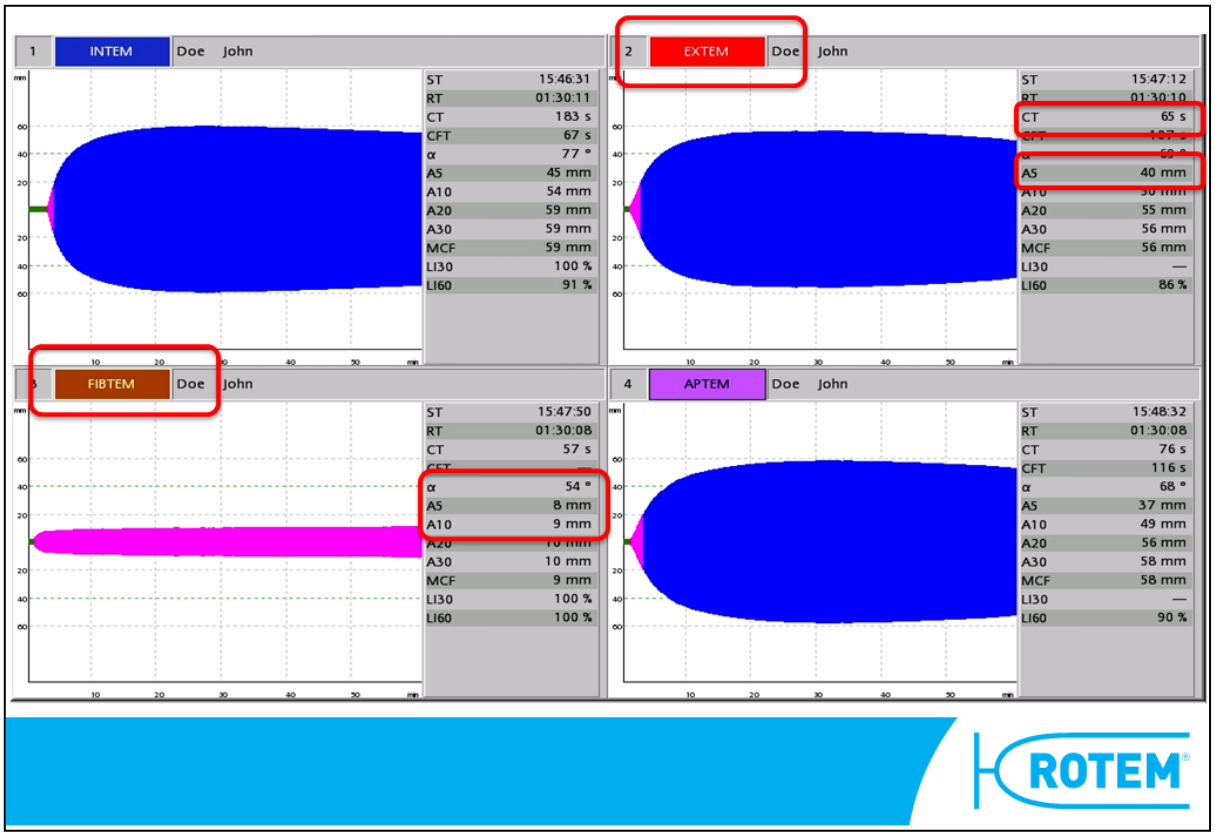


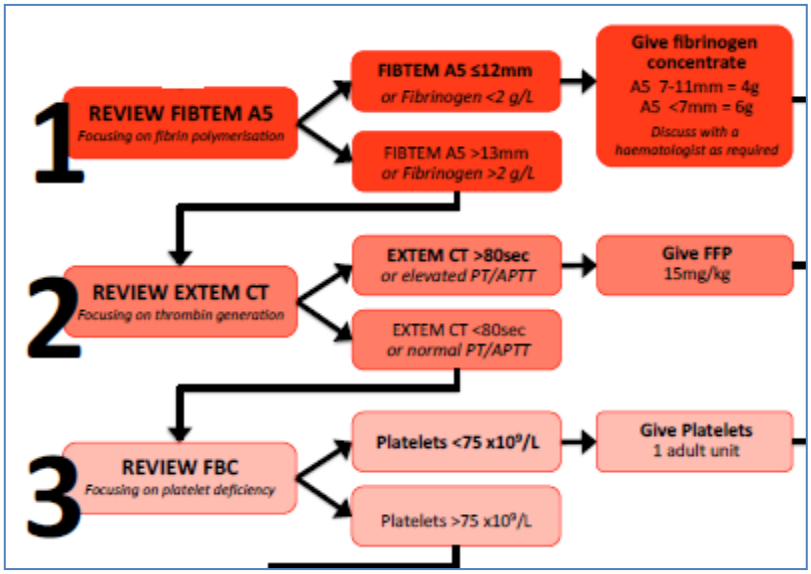
Figure 3: ROTEM Result Page

How do I find out more?

- 'ROTEM data interpretation teaching' pdf in the ROTEM folder in the FRCA drive.
- Obs Cymru Project: http://www.1000livesplus.wales.nhs.uk/obs-cymru
- Ask Gareth Roberts or Stephan Clements

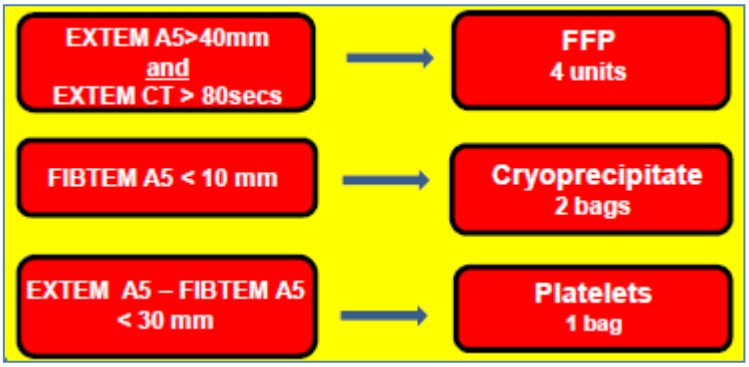
Appendix 1: Obstetric ROTEM Algorithm

N/B: This is for familiarisation only. Not to be used without referring to the full Obstetric Major Haemorrhage protocol.



Appendix 2: Non-Obstetric ROTEM Algorithm

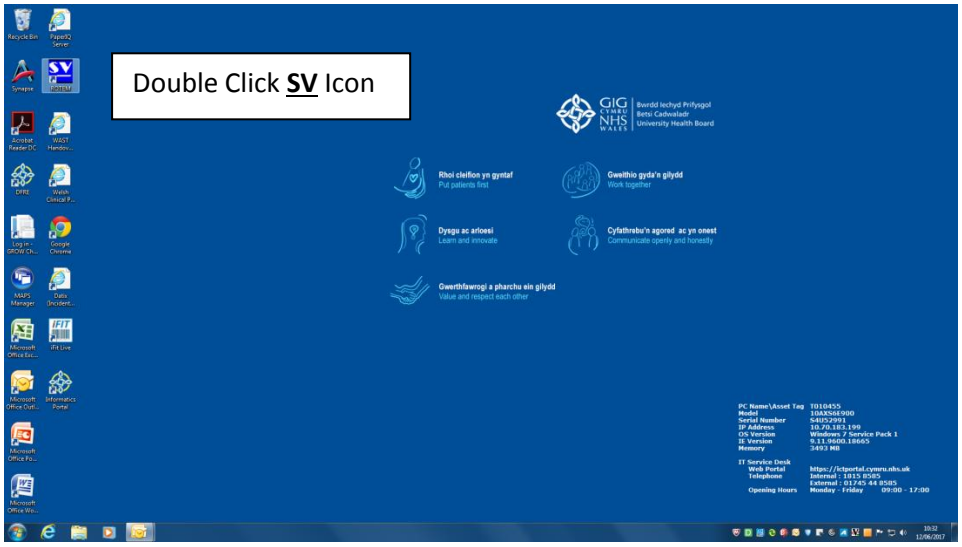
N/B: This is for familiarisation only. Not to be used without referring to the full Non-Obstetric Major Haemorrhage protocol.



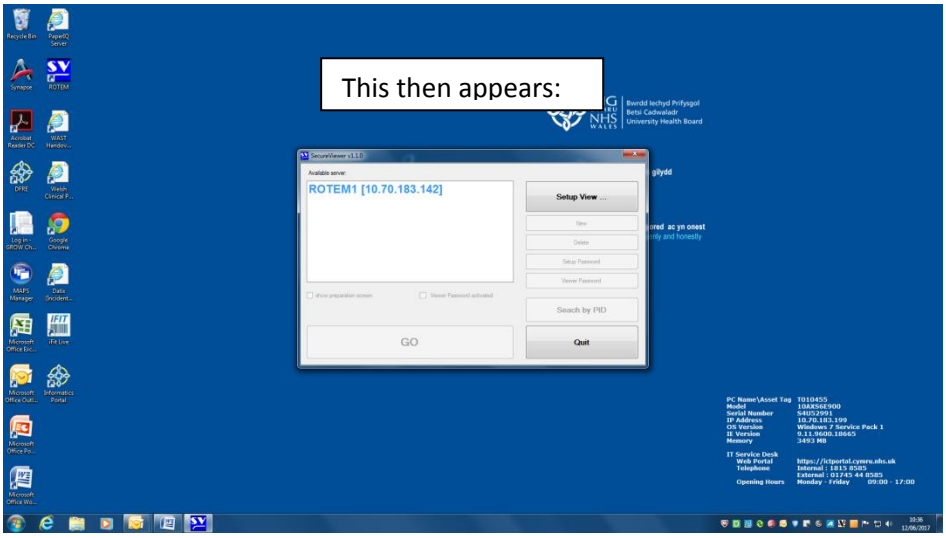
Appendix 3: Using Secure Viewer

The program, Secure Viewer allows you to remotely view the results of your ROTEM test. It can be accessed on the following computers: All main theatres, Cybi, Labour ward and ED Resus.

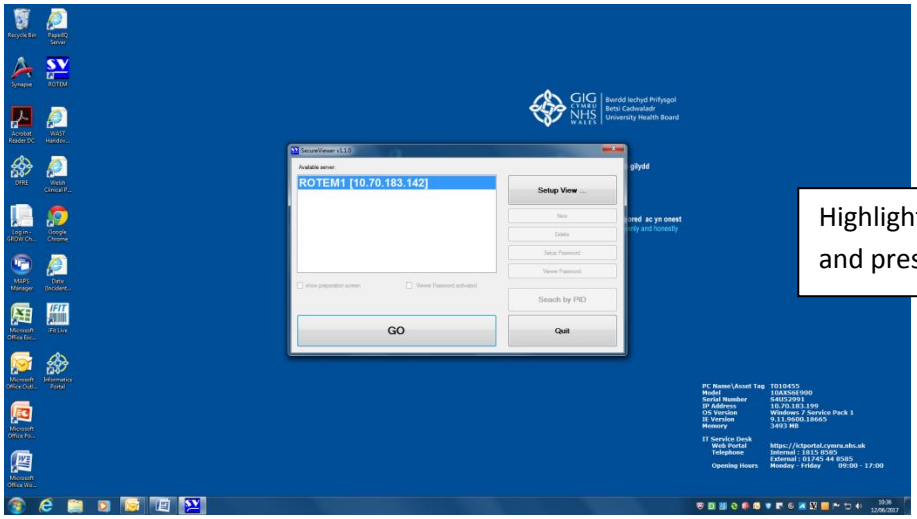
1.



2.

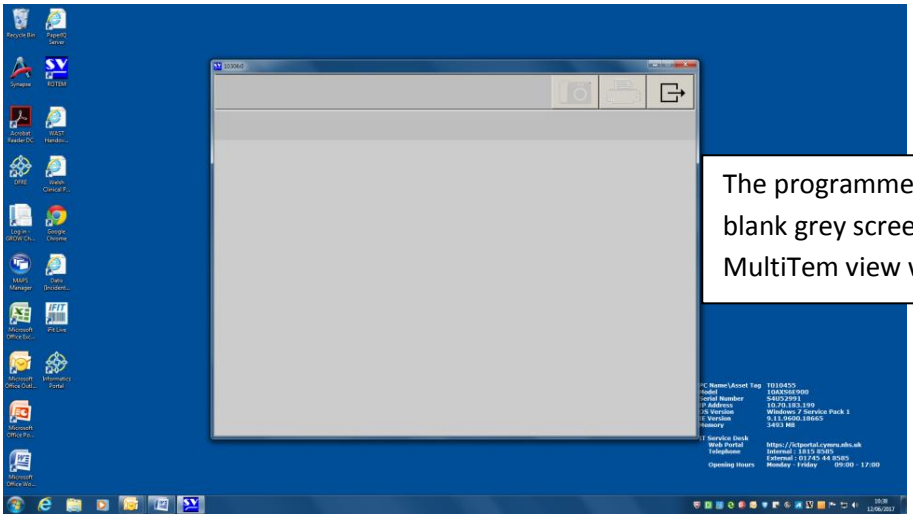


3.



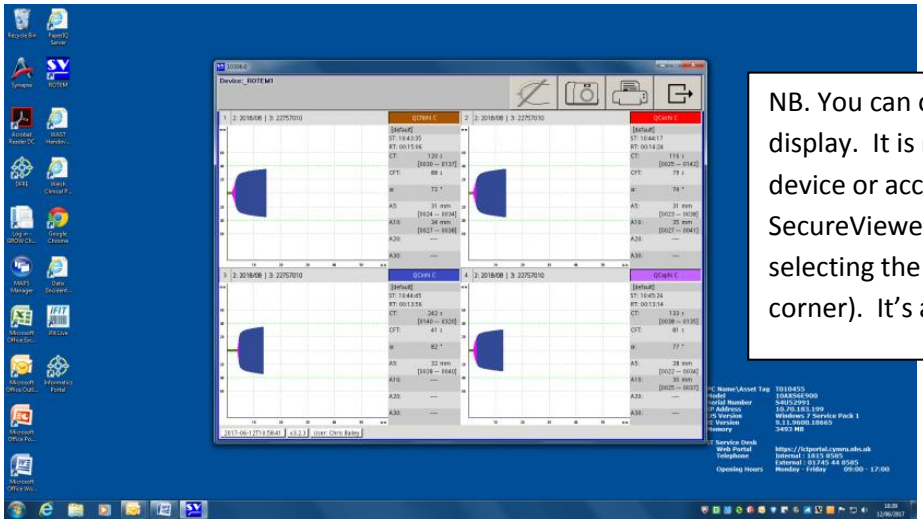
Highlight the server as indicated and press go.

4.



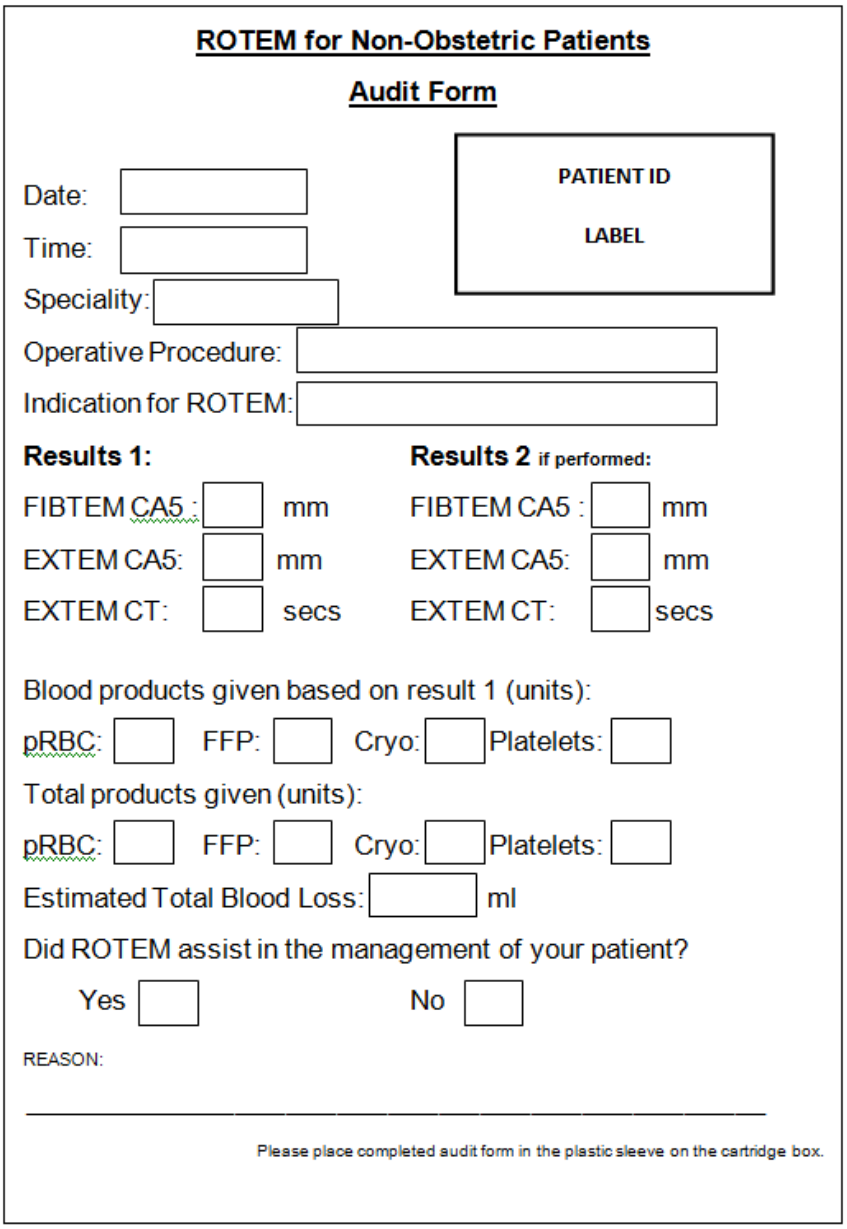
The programme may spend a few seconds with a blank grey screen. Please be patient and the MultiTem view will then open.

5.



NB. You can only view what is on the ROTEM display. It is not possible to manipulate the device or access its memory from SecureViewer. It is possible to print by selecting the print function (top right corner). It's a print screen function only.

Appendix 4: Non-obstetric use Audit form



ROTEM for Non-Obstetric Patients

Audit Form

PATIENT ID

LABEL

Date:

Time:

Speciality:

Operative Procedure:

Indication for ROTEM:

Results 1:		**Results 2** if performed:	
FIBTEM CA5 :	mm	FIBTEM CA5 :	mm
EXTEM CA5:	mm	EXTEM CA5:	mm
EXTEM CT:	secs	EXTEM CT:	secs

Blood products given based on result 1 (units):

pRBC: FFP: Cryo: Platelets:

Total products given (units):

pRBC: FFP: Cryo: Platelets:

Estimated Total Blood Loss: ml

Did ROTEM assist in the management of your patient?

Yes No

REASON:

Please place completed audit form in the plastic sleeve on the cartridge box.

Dr Stephan Clements

September 2017